.

Graduation Projects - Cloud Computing 2026/2027

Graduation Projects Proposal:

- **Total projects : 20 projects**
- **Number of Students per project: 4 Students per project (Group)**
- **Project Group: Mix of Credit Hours Students and Main Stream Students**
- **Background: Basic knoweldge of Distrubted systems and could computing**
- **The projects implemtion will be over AWS**

1- AI-Generated response and decision making solution for cloud operations

< project target: build an Cloud-AI chatbot system to perform incident management, resource monitoring, cost optimization, and troubleshooting.

The chatbot will use AI to interpret queries, provide insights, and automate decisions or trigger actions within the cloud environment.

2- Implementing AWS Server-based and Serverless Application Model

< project target: build scalable, cost-efficient, and easily manageable cloud applications by implementing Server-based applications give you more control over the infrastructure, while serverless applications using AWS SAM provide convenience and scalability with minimal operational overhead.

3-Three-tier Cloud Application Deployment with Identity management backend system using AWS

< project target: implement three layers: Presentation Layer, Application Layer, and Data Layer. Additionally, identity management will handle user authentication, authorization, and user data security.

4- Metrics Aggregation solution for Kubernetes Clusters

< project target: collect, aggregate, and visualize metrics from your Kubernetes clusters using Prometheus and Grafana, with alerting and long-term storage via Thanos tool

5- Automated AWS Infrastructure Provisioning

< project target: Using Ansible for AWS infrastructure provisioning allows you to easily manage and scale your AWS resources without needing to manually create and configure each one. By writing YAML playbooks, you can automate tasks such as provisioning EC2 instances, creating S3 buckets, setting up security groups, and managing RDS instances, reducing manual intervention and improving the efficiency of your workflows.

6- Kubernetes vertical and horizontal Autoscaling with Custom resources metrics

< project target: implementing Kubernetes Horizontal Pod Autoscaler (HPA) and Vertical Pod Autoscaler (VPA), you can scale your applications both horizontally (number of pods) and vertically (resource requests and limits). Custom metrics will be used in scaling based on application-specific requirements, like request rate, queue length, or other business metrics.

7- Auto Scaling for Stateful Applications with persistent storage

< project target: Autoscaling for stateful applications in Kubernetes with persistent storage can be challenging because stateful applications (such as databases or message queues) typically require persistent storage and consistent identity for each pod. However, Kubernetes provides solutions for both horizontal and vertical auto scaling of stateful applications while ensuring the persistence of storage.

8- Autoscaling for Serverless Frameworks using KNative

< project target: Knative provides an efficient way to implement serverless autoscaling with Kubernetes. expected to use Knative Serving and Knative Eventing, you can autoscale both stateless and stateful applications based on HTTP traffic or events.

9- Kubernetes Autoscaling with Machine Learning based on traffic load prediction

< project target: integrating machine learning for traffic load prediction, you can forecast future workloads and scale the cluster proactively

10- Implementing High Availability solution for kubernetes Clusters

< project target: Kubernetes workloads remain available and performant even in the event of hardware or software failures, that is accomplished by configuring HA for control nodes, workers, application, storage and networking.

11- Onboarding virtual networking function (VNF) using Openstack private cloud infrastructure

< project target: VNFs are network functions (e.g., routers, firewalls, load balancers, telco nodes) that traditionally ran on physical appliances but are now virtualized and deployed on cloud infrastructure, in this project you will build openstack environment and integrate MANO orchestration layer with openstack NFVI for orchestrating VNF lifecycle

12- Kubernetes Self-Healing and automation deployment using GitOps

< project target: build kubernetes self-healing capabilities, automated deployments, and ensures that your Kubernetes cluster consistently matches the desired state defined in your Git repository.

13- Deploying container-based applications using Helm charts in kubernetes cluster

< project target: Helm charts in a Kubernetes cluster simplifies and automates the deployment process .

Helm is a package manager for Kubernetes that helps in defining, installing, and upgrading Kubernetes applications. Helm charts contain all the necessary configuration for deploying applications, including Kubernetes resources like deployments, services, and ingress.

14- Create a distributed object storage solution with Kubernetes and MinIO

< project target: Deploy a highly available, distributed object storage system using Kubernetes to replicate data across multiple kubernetes nodes. MinIO provides S3-compatible storage, which is great for managing backups or large datasets in a scalable way.

15- Cloud instances monitoring using prometheus

< project target: monitor your AWS EC2 instances and other AWS resources using Prometheus and CloudWatch Exporter. Prometheus will collect the metrics, and you can visualize them using Grafana for real-time monitoring. Additionally, you will set up alerting to be notified when critical thresholds are breached.

16- High availability solution across multiple Availability Zone with self-healing for AWS EC2 instances

< project target: in this project you will focus on Deploying EC2 instances across multiple AZs, Using an Elastic Load Balancer (ALB or NLB) to distribute traffic, Automating scaling and healing with Auto Scaling Groups (ASG) and CloudWatch Alarms, Ensuring self-healing by replacing unhealthy instances.

17-Disaster Recovery Automation solution using Ansible

< project target: implement a Disaster recovery plan including automating backup, failover, restoration, testing, and validation, this minimizes downtime and quickly restore services to normal operations after a disaster.

18- Hybrid Workload Orchestration solution for Containers and VMs

< project target: integrating Kubernetes and KubeVirt to manage both containerized applications (pods) and virtual machines (VMs) within a single Kubernetes cluster.

This enables a hybrid approach where you can run modern cloud-native workloads (containers) alongside traditional VM-based workloads, providing flexibility and efficient orchestration.

19- Implementing Continuous Integration and delivery (CI/CD) Pipeline with Jenkins and AWS EC2

< project target: automate the process of building, testing, and deploying your applications to production. Jenkins is a popular automation tool that integrates with various services, including AWS EC2, to streamline the software delivery process.

20- Kubernetes Cluster Autoscaler (CA) using AWS

> project target: automatically scale your Amazon EC2 instances based on the workload demands in your Kubernetes cluster. This helps you manage resource allocation efficiently, reduce costs by scaling down unused resources, and ensure that your applications have the required resources when needed.

END of Exam, Good Luck

Projects Supervisor
Dr. Wagdy Anis Aziz

Exam. Date:12th of Aug., 2026